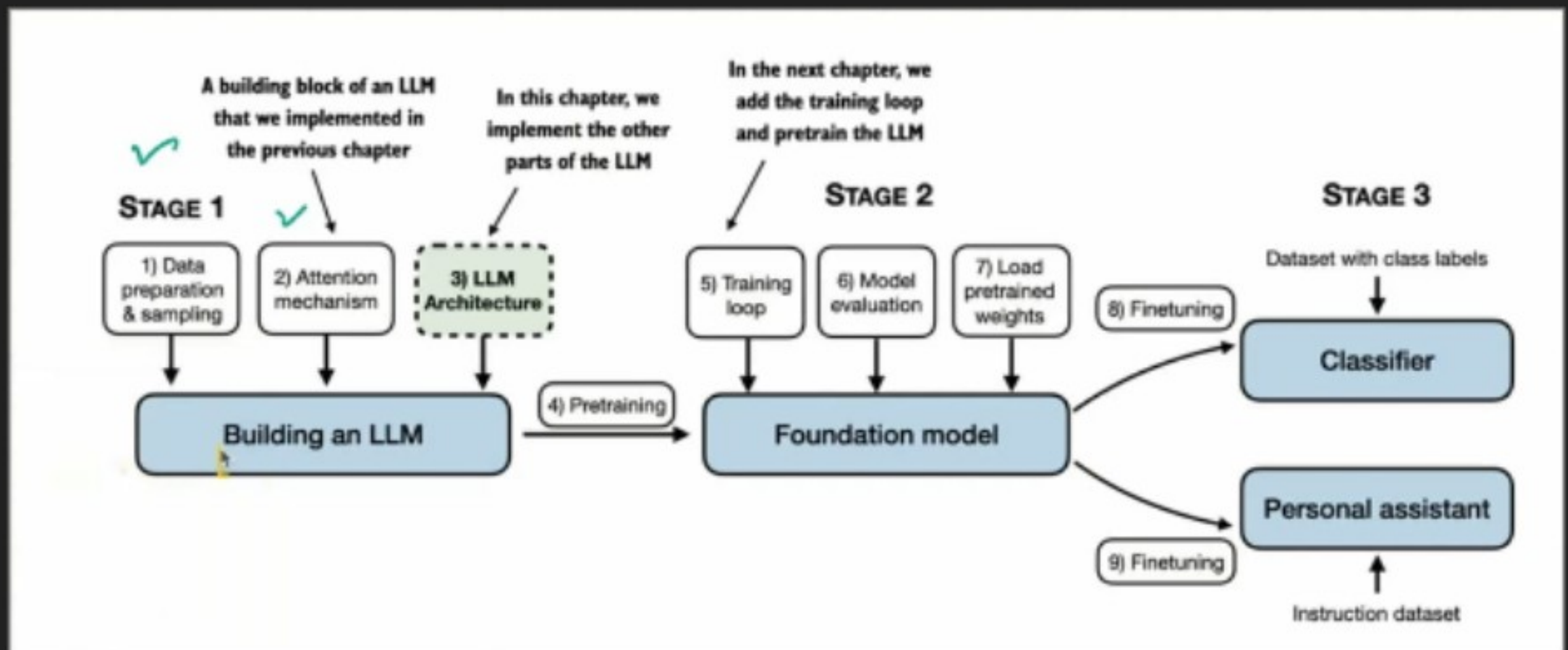
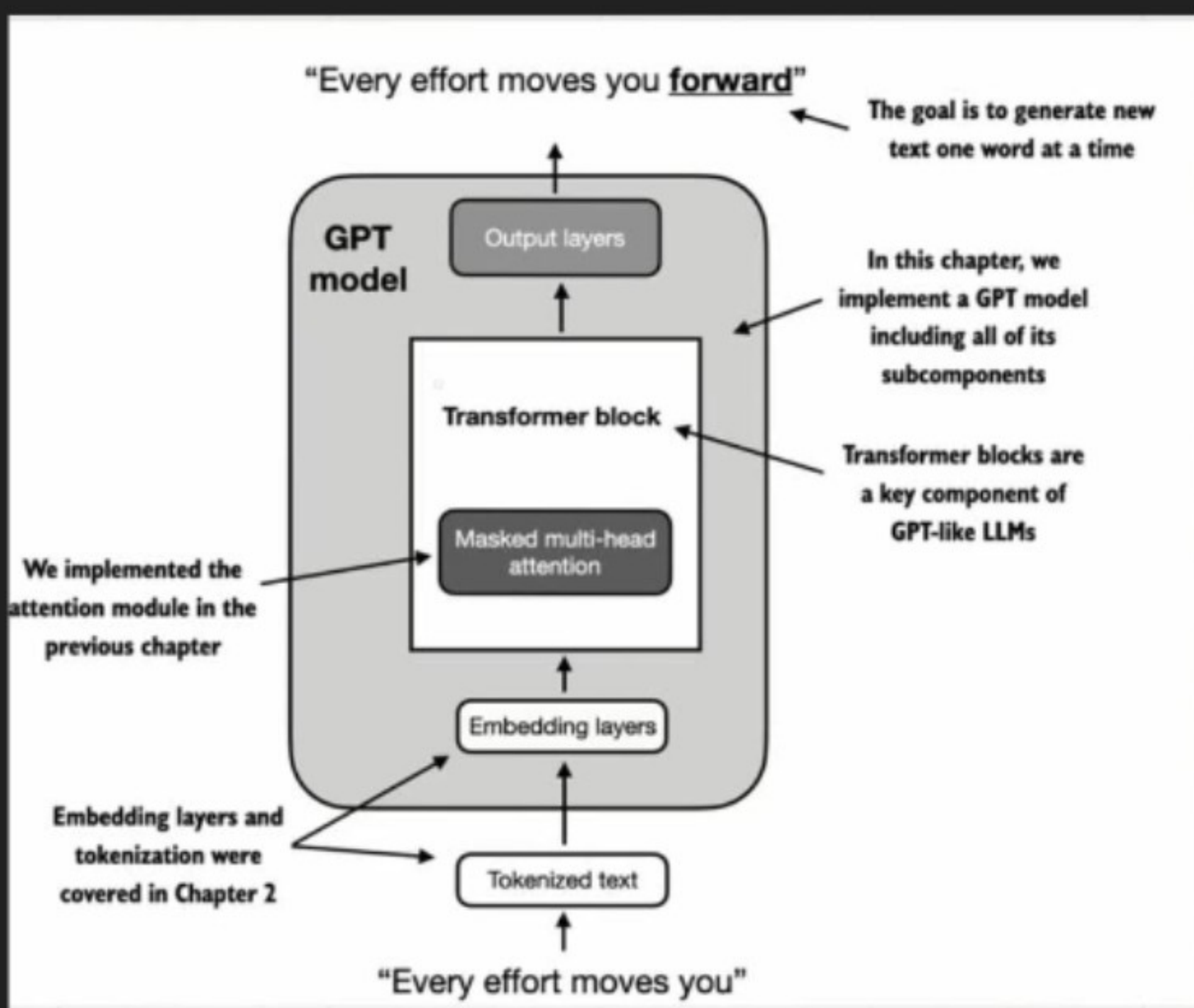


Lecture - 19: LLM Architecture Overview



After learning about the attention mechanism let's us learn about the LLM architecture now:

① coding on LLM architecture



Zoom in Transformer block

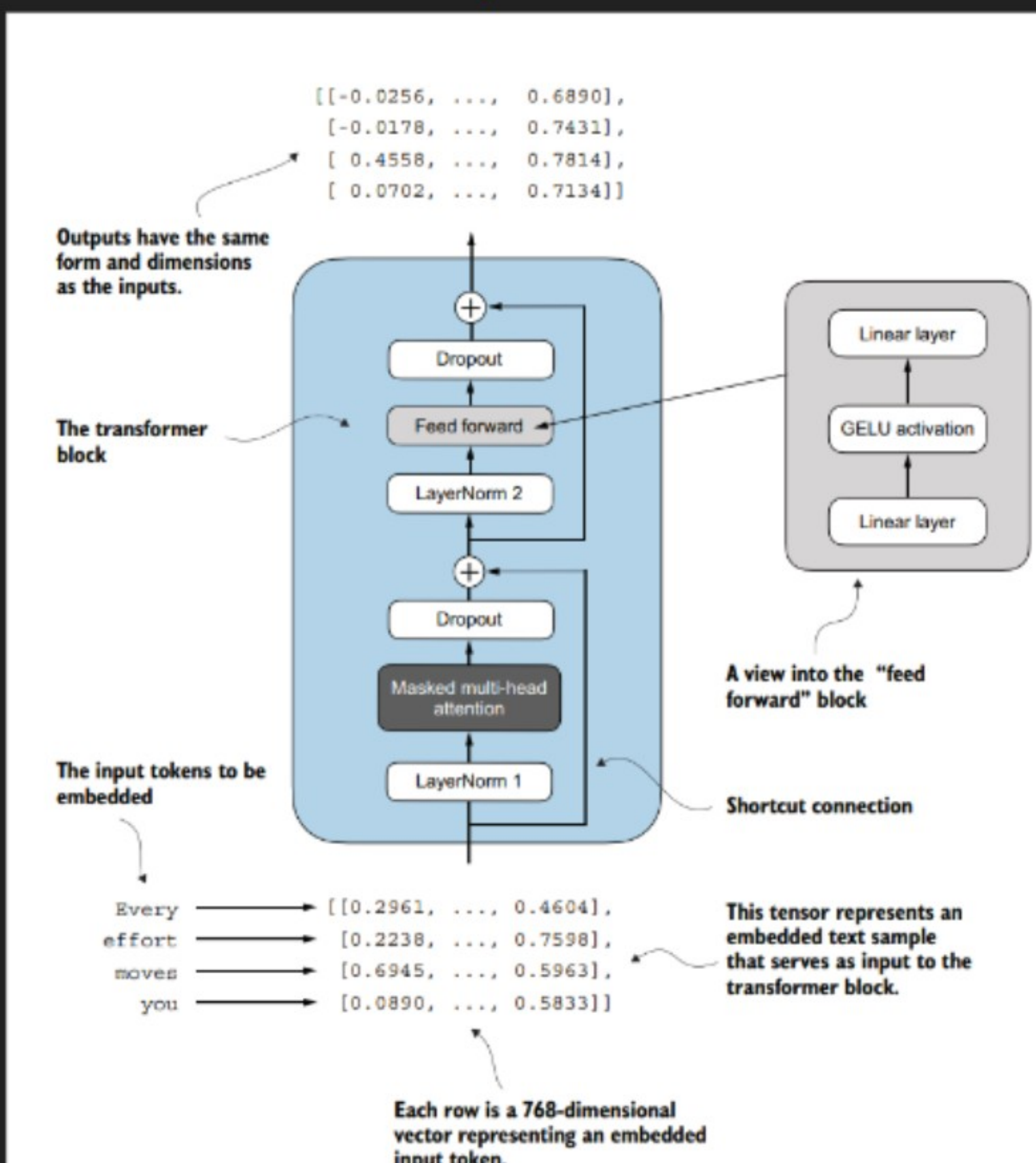


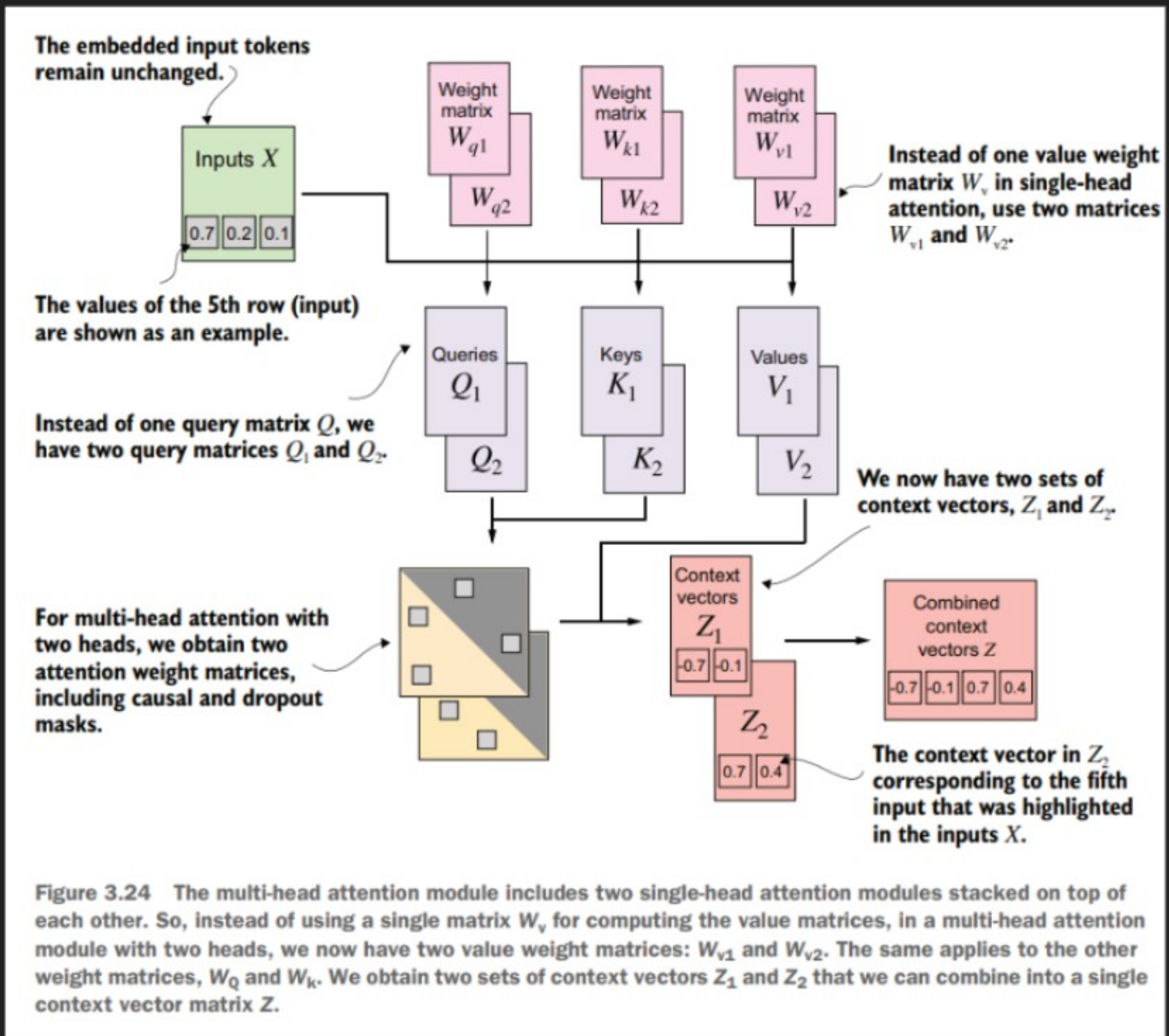
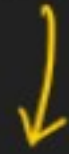
Figure 4.13 An illustration of a transformer block. Input tokens have been embedded into 768-dimensional vectors. Each row corresponds to one token's vector representation. The outputs of the transformer block are vectors of the same dimension as the input, which can then be fed into subsequent layers in an LLM.

② what we have learnt so far:

① Input tokenization

② Embedding (Token + positional)

③ masked multi-head attention



③ what we are yet to learn:

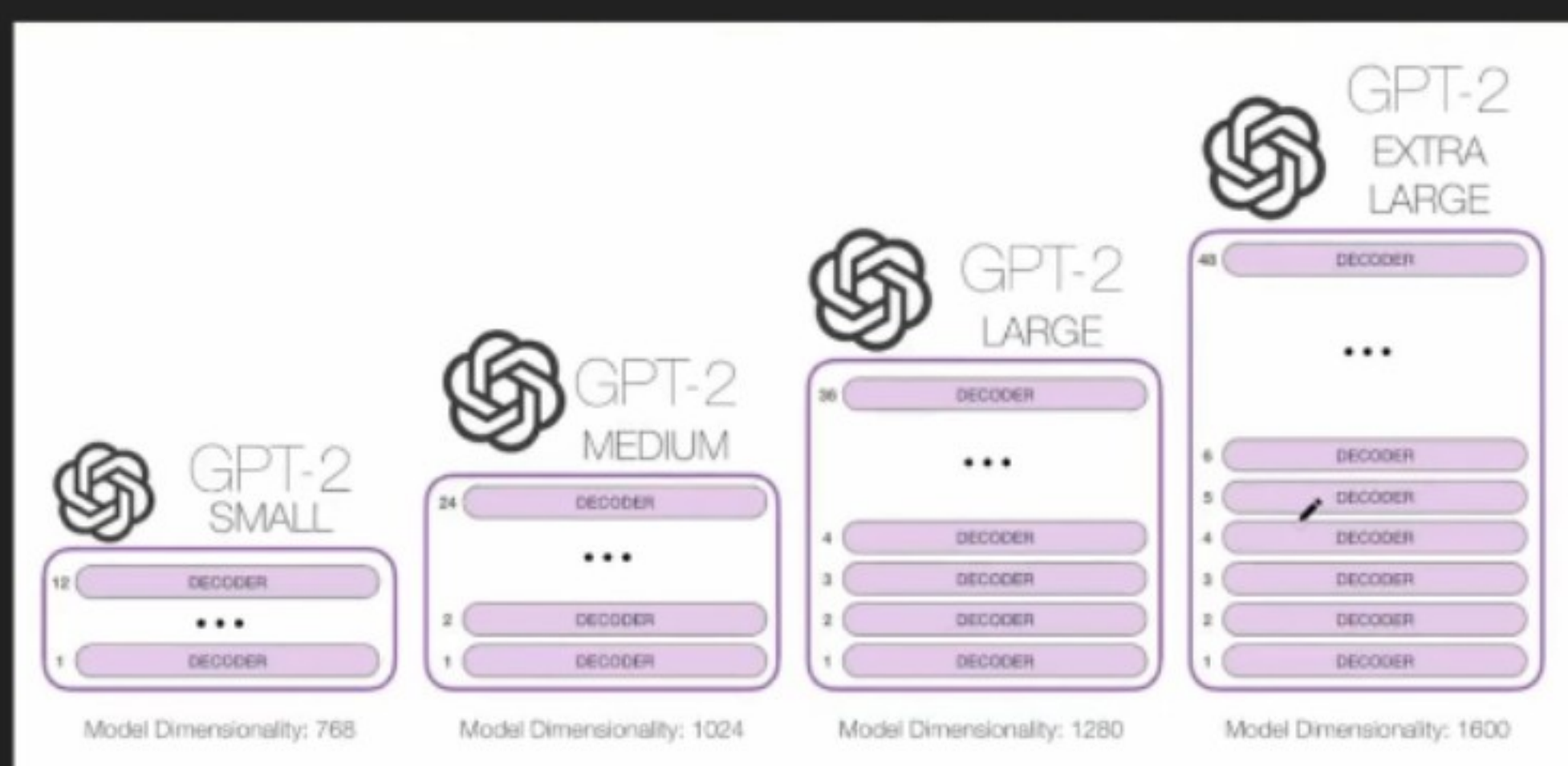
① Transformer blocks

④ we will scale up to the size of a small GPT-2 model → 124 million parameters.

⑤ Open AI has made GPT-2 weight public. GPT 3.4 weight have not yet been made public

$GPT - config - 124M = \{$
 $Vocab - size = 50257,$
 $Context - length = 1024,$
 $emb - dim = 768,$
 $n - heads = 12,$
 $n - layers = 12,$
 $drop - rates = 0.1,$
 $qkv - bias = false;$
 $\}$

Number of
transformer
blocks



using this configuration, we will build
a GPT placeholder architecture.

(Dummy GPT model)

This will give us a birds eye view
of how everything fits together.

